

MO-IT117 Data Visualization Techniques

Data Cleaning and Insights Report

GROUP DETAILS					
Group Name/No.	Group 3				
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PROJECT OVERVIEW	
Dataset Name:	Philippine Statistics Authority (PSA) 2024 Internet Use and ICT-Related Regional Datasets
Source:	Philippine Statistics Authority (PSA)
Project Focus (Domain/Context):	Information and Communications Technology (ICT) and Internet Usage in the Philippines, focusing on regional differences in internet activities, demographics, frequency of use, and network/device conditions.
Analytical Purpose:	To present the state of digital connectivity across Philippine regions by looking at internet usage, device and network connectivity conditions, and demographic patterns among individuals aged 10 years and over. The goal is to identify areas where connectivity gaps still exist and use the findings to suggest possible improvements for present and future digital infrastructure, accessibility, and inclusion.
Target Audience:	A mixed audience consisting of ICT professionals, government and community decision-makers, students, and general audiences interested in digital technology and internet usage in the Philippines. The visualizations will be designed to be understandable to users with different levels of data expertise.

KEY VARIABLES FOR ANALYSIS	Variable	Why It Matters
Identify the main variables you will use.	Region	Identifies the Philippine region being analyzed and allows comparisons of internet usage and ICT conditions across locations.
	Internet Activity	Shows the different ways people use the internet, such as social networking, email, streaming, online learning, and other activities.
	Number	Represents the number of people associated with each internet-use category and provides the basis for comparing the size of each group.
	Percentage	Shows the proportion associated with each category and allows easier comparison between regions and groups.
	Gender	Allows comparison of internet use between males and females across Philippine regions.
	Age Group	Shows how internet use varies among different age groups.
	Frequency	Shows how regularly people use the internet, including daily, weekly, monthly, and less regular use.
	Network / Device / Signal	Provides information about working devices, active communication capability, and 2G, 3G, 4G, and 5G network signals across regions.

DATA ISSUES and CLEANING DECISIONS	Field	Issue Identified	Action Taken (Exclude / Fix / Retain)	Rationale
For each issue, describe: what problem you observed what action you took why you chose that action	Header rows	The CSV files contained title/header information before the actual column headers and data.	Fixed	Tableau Prep was configured to use the correct header row and data starting row so the dataset would be interpreted correctly.
	Source Row Number	Tableau Prep added a technical source-row identifier that was not part of the original analytical variables.	Excluded	It was not needed for analysis and could create an unnecessary field in the cleaned datasets.
	Number and Percentage fields	Number and Percentage measures were stored in separate columns for each category.	Fixed	The fields were reshaped into a standardized structure with separate Number and Percentage measures, making the data easier to analyze and visualize.
	Internet activity fields	Internet activities were stored as multiple columns rather than as a single categorical field.	Fixed	Because these activities were scattered across separate files with redundant column structures, we pivoted the fields to eliminate overlap and consolidate them into a single file for cleaner output.
	Gender fields	Gender categories were stored across multiple measurement columns.	Fixed	Gender, Number, and Percentage fields were spread across multiple sources; we pivoted it into a master clean file to make the analysis more streamlined.
	Age group fields	Age groups were stored across multiple Number and Percentage columns.	Fixed	Same as the Gender fields, we consolidated and pivoted the demographic data to create a single file for cleaner presentation.
	Frequency fields	Internet-use frequency categories were stored across separate columns.	Fixed	Internet-use frequency was different across columns, so we combined and pivoted the columns into a master file to create a single, relational structure.
	Network and signal fields	Working-device and network-signal categories were stored in separate columns.	Fixed	Connection metrics were located in separate columns; we reshaped them into a Network / Device / Signal category with separate Number and Percentage measures.
	Missing values	The cleaned datasets were checked for null values in the main Number and Percentage fields. No missing values were found in the final analytical fields.	Retained	No records needed to be removed or replaced because the required analytical values were present.
	Duplicate records	Duplicate combinations were checked using the relevant category and Region fields. No duplicate analytical records were found.	Retained	The records were unique and therefore did not require deletion.
	Region	Region was the common geographic field across all five datasets.	Retained	Region is essential for comparing internet-use patterns across Philippine regions.
	Percentage values	Percentage values were numeric values using a 0–100 scale.	Retained	The original numeric values were preserved so they can be correctly formatted as percentages during visualization without changing the underlying data.
	Not Reported age group	The Age Group dataset contained a Not Reported category.	Retained	It represents information contained in the source dataset and was retained rather than removed.

RESULTING DATASET	
After cleaning: What changes were made to the dataset?	After cleaning the datasets, several checks were performed to ensure the data remained complete and accurate.
	First, the values in the cleaned datasets were compared with the original PSA files. All 810 cleaned records matched the original Number and Percentage values, confirming that the reshaping and renaming steps did not alter the data.
	The number of records was also checked against the expected number of regions and categories. The final datasets contained: - 396 records for Internet Use by Type - 54 records for Gender - 180 records for Age Group - 90 records for Frequency of Use - 90 records for Network, Device, and Signal
	No missing category combinations or duplicate Region and Category records were found.

<i>What data was excluded, fixed, or retained?</i>	The datasets were also compared with each other when they contained the same overall internet-use figures. The totals reported in the Gender, Age Group, and Frequency of Use datasets matched across all 18 geographic categories. This provided another check that the data was cleaned and organized correctly.
<i>Is the dataset now ready for analysis? Why or why not?</i>	Additional checks were made on the percentages. Male and Female percentages added up to 100%. Age Group and Frequency percentages were also close to 100%, with small differences caused by rounding in the original PSA data. The Network and Signal percentages were not expected to add up to 100% because some categories may overlap. Because of this, they were checked for valid values and consistency instead of treating them as parts of one total. The cleaning workflow used for these datasets can also be applied to future PSA regional datasets with a similar wide-format structure. The process includes identifying the correct header row, removing unnecessary fields, checking data types, identifying repeated Number and Percentage columns, reshaping the categories into rows, standardizing field names, checking for missing values and duplicates, comparing the cleaned values with the original data, and checking matching totals across related datasets. As a secondary validation measure, the aggregated population totals from our file were cross-checked against the official PSA regional population figures. This external analysis verified our internal values and confirmed that the numbers proportionally reflected the known demographic distribution across all 18 regions, confirming the dataset is accurate beyond internal 'null' and duplicate checks.

INSIGHTS FROM DATA CLEANING

<i>What did you discover while cleaning?</i>	The cleaning process revealed that all five datasets use the same 18 Philippine regional categories, providing a consistent geographic basis for comparison.
<i>Examples:</i>	The datasets also have different levels of categorization. Internet Use by Type is organized by internet activity, while the other datasets organize information by gender, age group, frequency, and network/device/signal conditions. Because these datasets have different structures and analytical categories, combining all of them into one physical table could create repeated records and affect aggregate calculations. They are therefore maintained as separate cleaned datasets and can be compared using the common Region field.
<i>patterns in missing values unusual or unexpected entries early trends or relationships</i>	The cleaning process also confirmed that Number and Percentage are available as separate measures. This provides two ways to examine the data: Number can show the size of the population represented, while Percentage can support comparisons between regions and categories.
<i>Explain how these insights may influence your visualizations.</i>	The Age Group dataset also contains a Not Reported category, which was retained rather than removed so that the source information remains complete. These findings will guide the later analysis toward regional comparisons and comparisons among internet activities, demographic groups, usage frequency, and network/device conditions. The cleaned structure also allows the same types of measures to be used consistently across the different datasets.